Random Experiment: An activity/process in which outcome is unknown.

ex) selecting student at random

Sample Space: The collection of all possible outcomes of a random experiment notation is Ω.

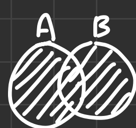
ex) $\Omega = \{1, 2, 3, 4, 5, 6\}$

Event: A subset of sample space

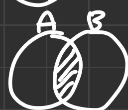
Outcome: Element of a sample space; result of a random experiment

$A \subseteq B$. A is a subset of B.

$A \cup B$ Union



$A \cap B$ Intersection



A^c Complement

De Morgan's Laws

$(A \cup B)^c = A^c \cap B^c$ $(A \cap B)^c = A^c \cup B^c$

Probability of an event is a measure of how likely an event to occur.

Axiom

1. For any event A, $P(A) \geq 0$
2. $P(S) = 1 \rightarrow P(\Omega) = 1$
3. If $A_1, A_2, A_3, \cdots,$ is infinite collection of disjoint events, then

$$P(A_1 \cup A_2 \cup A_3 \cup \cdots) = \sum_{i=1}^{\infty} P(A_i)$$

Addition Rule

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

If there are N equally-likely simple events (outcomes) in sample space Ω, and event A can be expressed as the unison of simple events E_i, then

$$P(A) = \sum_{E_i \text{ in } A} P(E_i) = \sum_{E_i \text{ in } A} \frac{1}{N} = \frac{N(A)}{N}$$

where $N(A)$ is the number of simple events in event A.

Conditional Probability

For any events A and B, with $P(B) > 0$, the conditional probability of A given B is

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad \text{OR} \quad P(A \cap B) = P(A|B)\,P(B)$$

Indepent Events : Events A and B are independent

if

$P(A|B) = P(A)$

$P(B|A) = P(B)$

$P(A\cap B) = P(A)P(B)$

(A can be replaced by A^c, B to B^c ...)

one of them are true, they are independent.

if not, they are <u>dependent</u>.

Inclusion - Exclusion Principle

For any events $A_1, A_2, \cdots, A_n$,

$$P(A_1 \cup A_2 \cup \cdots \cup A_n)$$

$$= \sum_{i=1}^{n} P(A_i) - \sum_{i<j} P(A_i \cap A_j) + \sum_{i<j<k} P(A_i \cap A_j \cap A_k) - \cdots$$

$$(-1)^{n+1} P(A_1 \cap A_2 \cap \cdots \cap A_n)$$

Counting · If we can do task 1 in w_1 ways and task 2 in w_2 ways, Then we can do either task 1 or task 2 (not both) in $w_1 + w_2$ ways.

- If we can do task 1 in w_1 ways and task 2 in w_2 ways, (unrelated). we can do both in $w_1 \times w_2$ ways.

FPC (Fundamental Principle of Counting)

Consider a collection of n ordered elements $(a_1, a_2, \cdots, a_n)$. If there are k_1 possible value for the first element, k_2 for second... and there are $k_1 \times k_2 \times \cdots \times k_n$ possible sequences of n elements.

Permutations "n Pick r" The number of ways we can arrange r objects from n distinct objects.

$$_n P_r = \frac{n!}{(n-r)!}$$

Combinations "n choose r" The number of ways to select r objects from n objects.

$$_n C_r = \frac{n!}{r!(n-r)!}$$

Mutually Exclusive (disjoint) if event A happens and B cannot happen, A and B are mutually exclusive.

Law of Total Probability

Suppose $B_1, \cdots, B_k$ is a partition of the sample space. Then,

$$P(A) = \sum_{i=1}^{k} P(A|B_i)P(B_i)$$

Bayes Formula

$$P(B|A) = \frac{\overset{P(A\cap B)}{P(A|B)P(B)}}{\underbrace{\underbrace{P(A|B)P(B)}_{P(A\cap B)} + \underbrace{P(A|B^c)P(B^c)}_{P(A\cap B^c)}}_{=P(A)}}$$

- Formula
- Axiom
- Term (Def)
- Math term

Random Variable

Let Ω be a sample space. A discrete random variable is a function $X: \Omega \to \mathbb{R}$ that takes on a finite number of values $a_1, a_2, \cdots, a_n$ or a countably infinite number of values $a_1, a_2, \cdots$

PMF [Probability Mass Function]

PMF of a discrete random variable, X,

is one that assigns a probability to each value x s.t.

$$0 \le P(X=x) \le 1 \text{ for all } x$$

AND

$$\sum_{\text{all } x} P(X=x) = 1$$

* Rules related to PMF

Assume $P(X=0)=0.3$
$P(X=1)=0.2$
$P(X=2)=0.5$

then

$$\begin{aligned} &P(X>0) \\ &= P(X \ge 1) \\ &= P(X=1)+P(X=2) \\ &= p(1)+p(2) \\ &= 0.7 \end{aligned}$$

CDF [Cumulative Distribution Function]

Let X be a random variable.

The CDF of x is a function

$$F(x) = P(X=x)$$

defined for all real numbers x.

↳ the function F(x) is a

- non-decreasing function
- step function
- right-continuous function

- bounded by 0 and 1

Expectation (Mean)

If X is a discrete random variable that takes values in a set S, its expectation, $E[X]$ is defined as

$$E[X] = \sum_{x \in S} x P(X = x)$$

Expected Value of X is often denoted as μ or μ_X

Expectation of function of a Random Variable

- Let X be a random variable that takes values in a set S. Let g be a function. Then,

$$E[g(X)] = \sum_{x \in S} g(x) P(X = x)$$

- for constants a, b, and a random variable X with expectation $E[X]$.

$$E[aX + b] = aE[X] + b$$

Variance

Let X be a random variable with mean $E[X] = \mu < \infty$. The variance of X is

$$V[X] = E[\underbrace{(X-\mu)^2}_{g(X)}] = \sum_x \underbrace{(x-\mu)^2}_{g(x)} \underbrace{P(X=x)}_{p(x)} \quad \overset{(=\sigma^2)}{\sigma_x^2}$$

The standard derivation of X is

$$SD[X] = \sqrt{V[X]}$$ — same unit as X. $\overset{(=\sigma)}{\sigma_x}$

Computational Formula for Variance

$$V[X] = E[X^2] - E[X]^2$$

Proof.

$$V[X] = E[(X-E(X))^2]$$
$$= E[X^2 - 2X\underbrace{E(X)}_{\mu} + \underbrace{(E(X))^2}_{\mu}]$$

Let $y = g(X) = X^2$

$$\left.\begin{aligned} b &= 2E(X) \\ c &= [E(X)]^2 \end{aligned}\right\} \text{constants}$$

$$\because E[y - bX + c]$$
$$= E(y) - bE(X) + [E(X)]^2$$
$$= E(X^2) - 2E(X)[E(X)] + [E(X)]^2$$

$$= E(X^2) - [E(X)]^2$$

- Properties of Variance

$$V[aX+b] = a^2 V[X]$$

* X and Y are independent

$$V[X+Y] = V[X] + V[Y]$$

- Formula
- Axiom
- Term (Def)
- Math term

Markov's Inequality

Limits how likely extremely large values are, using mean

Let X be a non-negative random variable and let constant $a > 0$. Then,

$$P(X \geq a) \leq \frac{E(X)}{a}$$

Proof.

$$E(X) = \sum_{\text{all } x} x p(x)$$

$$= \underbrace{\sum_{x<a} x p(x)}_{>0} + \sum_{x \geq a} x p(x)$$

$$\geq \sum_{x \geq a} x p(x) \geq \sum_{x \geq a} a p(x) = a \sum_{x \geq a} p(x)$$

$$= aP(X \geq a)$$

$$\therefore E(X) \geq aP(X \geq a) \rightarrow P(X \geq a) \leq \frac{E(X)}{a}$$

Chebyshev's Inequality

limits how likely values far from mean are, using variance

Let X be a random variable with mean μ and finite variance σ^2. Then, for any $k>0$,

$$P(|X-\underset{E(X)}{\mu}|<k\underset{\sqrt{V(X)}}{\sigma}) \geq 1-\frac{1}{k^2}$$

probability X is within $k\sigma$ within $\mu=E(X)$

$\Rightarrow P(\mu-k\sigma<X<\mu+k\sigma)$

Proof.

$$P(|X-\mu|<k\sigma) = P[\underbrace{(X-\mu)^2}_{\geq 0}<\underbrace{k^2\sigma^2}_{>0}]$$

$$=1-P[(X-\mu)^2 \geq k^2\sigma^2]$$

$$=1-\underbrace{P[y \geq a]}$$

use markov's ineq.

$$\leq \frac{E(y)}{a} = \frac{E[(X-\mu)^2]}{k^2\sigma^2} = \frac{\cancel{V(X)}^{\sigma^2}}{k^2\cancel{\sigma^2}} = \frac{1}{k^2}$$

$$\geq 1-\frac{1}{k^2}$$

$\therefore P(|X-\mu|<k\sigma) \geq 1-\frac{1}{k^2}$

and $P(|X-\mu| \geq k\sigma) \leq \frac{1}{k^2}$

Common Discrete Probability Distribution

- Bernoulli Distribution

Used to model a random experiment with two possible outcomes (dichotomous trial) where outcomes are labelled as :

- Success (outcome of interest)
- Failure (the complement of above)

Here, the random variable is an Indicator Variable where $X \begin{cases} 1, \text{success} \\ 0, \text{failure} \end{cases}$.

probabilities are $\begin{cases} \text{'success'} = p \\ \text{'failure'} = 1-p \text{ (sometimes } q) \end{cases}$

※ The PMF of bernoulli random variable is

$$p(x) = p^x (1-p)^{1-x}, \quad x = 0, 1$$

Binomial Experiment

An experiment consists of a sequence of smaller trials and has these properties:

1. fixed number of trials (n)
2. independent trials
3. dichotomous trials → success/failure
4. constant probability of success, p among trials

Binomial Distribution

A random variable X have a binomial distribution with parameters n and p if

$$P(X=k) = \binom{n}{k} p^k (1-p)^{n-k}, \text{ for } k = 0, 1, \cdots, n.$$

We write $X \sim \text{Binom}(n, p)$
or $\text{Bin}(n, p)$

$E(X) = np \quad Var(X) = np(1-p)$

R
x size prob
dbinom(k, n, p)
exact: $P(X=k)$
pbinom(k, n, p)
cumulative: $P(X \leq k)$

Poisson Experiment

- Describes the chances a random outcome will occur in an interval, (time or space) assuming that
  - The probability the outcome occurs is very small, but...
  - The # of trials is very large so that the event is bound to happen sometimes anyway.

Poisson Distribution

A random variable X has a poisson distribution with parameter $\lambda > 0$ if

$$P(X=k) = \frac{e^{-\lambda}\lambda^k}{k!}, \text{ for } k=0,1,\cdots,$$

$E(X) = \lambda$
$Var(X) = \lambda$

We write $X \sim Pois(\lambda)$

R
x lambda
dpois(k, λ)
ppois(k, λ)
cumulative

Hypergeometric Experiment

Samples n individuals/objects (n is fixed) from a set of N individuals/objects where:

1. N is finite
2. Each individual/object can be categorized as success/failure
3. r successes in population
4. Sampling without replacement, each set of n individuals equally likely to be chosen

Hypergeometric Distribution

A random variable X has the hypergeometric distribution with parameters r, N, λ if

$$P(X=k) = \frac{\overbrace{\binom{r}{k}}^{\text{success}} \overbrace{\binom{N-r}{n-k}}^{\text{failure}}}{\underbrace{\binom{N}{n}}_{\text{total}}}, \quad E(X) = n\frac{r}{N}, \quad Var(X) = n\left(\frac{N-n}{N-1}\right)\frac{r}{N}\left(1-\frac{r}{N}\right)$$

for $\max(0, n-(N-r)) \le k \le \min(n, r)$.

By domain of binomial coefficients,

$$0 \le k \le r, \quad 0 \le n-k \le N-r.$$

We write

$$X \sim HyperGeo(r, N, n)$$

$\mathcal{R}$

\# of choice

dhyper(k, r, N-r, n)   (x, m, n, k)

phyper(k, r, N-r, n)

cumulative

Geometric Experiment

Sequence of bernoulli trials (n not fixed) that continue until the first success is found.

1. independent trials
2. dichotomous trials (success / failure)
3. constant p across trials

~~fixed number of trials~~ → difference between binomial experiment

Geometric Distribution

A random variable X has the Geometric Distribution with parameter p if

$$P(X=k) = p(1-p)^k, \quad \text{for } k=0,1,2\cdots$$

We write $X \sim Geo(p)$.

$$E(X) = \frac{1-p}{p}, \quad Var(X) = \frac{1-p}{p^2}$$

R dgeom(k, p) (x: k, prob: p)

pgeom(k, p)

$*P(X \leq k) = 1-(1-p)^{k+1}$

Negative Binomial Experiment

Sequence of Bernoulli trials (n not fixed) that continue until the r^{th} success is observed.

1. independent trials
2. dichotomous trials (success / failure)
3. constant p across trials

Negative Binomial Distribution

Let X = total # of failures before r^{th} success.
Random Variable X has a Negative Binomial Distribution with parameters r and p if

$$P(X=k) = \binom{k+r-1}{r-1} p^r (1-p)^k, \quad k=0,1,2,\cdots$$

We Write $X \sim NB(r,p)$

$$E(X) = \frac{r(1-p)}{p} \quad Var(X) = \frac{r(1-p)}{p^2}$$

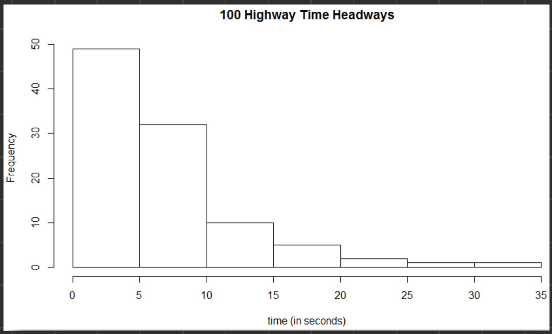
R dnbinom(k, r, p) (x: k, size: r, prob: p)

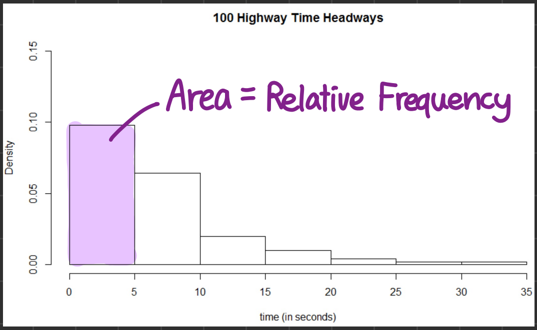
pnbinom(k, r, p)

⚠ Note: Have to add 1 to the $E(X) = \frac{r(1-p)}{p}$ if we want to include r^{th} success itself.

Histogram

Frequency vs. Density



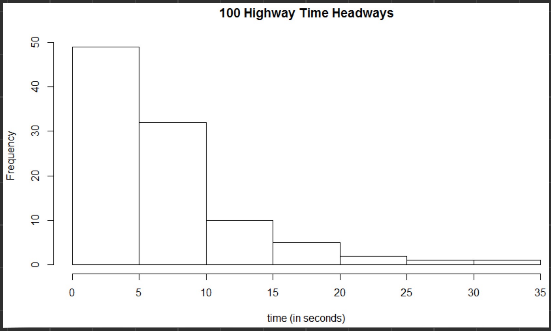


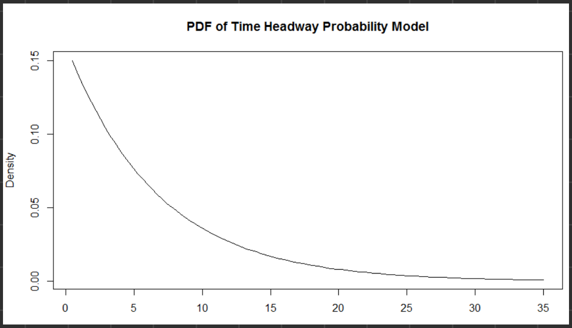
- **Frequency** Number of observations in each category

  Relative Frequency Proportion of observations in each category

$$\text{Density} = \frac{\text{Relative Frequency}}{\text{Width of Category}}$$

Data vs. Probability Model





- Relative frequencies of the 100 observed headways expressed as densities

- Theoretical Probabilities expressed as densities

Continuous Probability Distribution

Probability Density Function

Let X be a continuous random variable.
A function f is a probability density function of X if

1. $f(x) \geq 0$, for all $-\infty < x < \infty$
2. $\int_{-\infty}^{\infty} f(x)\,dx = 1$
3. For $S \subseteq \mathbb{R}$, $P(X \in S) = \int_S f(x)\,dx$

Cumulative Distribution Function

A function F is a cdf, that is, there exists a random variable X whose cdf is F, if it satisfies the following properties.

1. $\lim_{x \to -\infty} F(x) = 0$
2. $\lim_{x \to \infty} F(x) = 1$
3. If $a \leq b$, then $F(a) \leq F(b)$
4. F is right-continuous.

$\left(\lim_{x \to a^+} F(x) = F(a), \text{ for all } a \right)$

in q's, check for points where eq. changes

Relationship between PDF and CDF

Between the distribution function F and probability density function f of a continuous random variable,

$$F(b) = \int_{-\infty}^{b} f(x)\,dx \quad \text{and} \quad f(x) = \frac{d}{dx} F(x)$$

Expectation and Variance for C.R.V

For random variable X with density function f,

$$E[X] = \int_{-\infty}^{\infty} x f(x)\,dx$$

$$V[X] = \int_{-\infty}^{\infty} (x - E[X])^2 f(x)\,dx$$

If X has density function f, and g is a function, then,

$$E[g(X)] = \int_{-\infty}^{\infty} g(x) f(x)\,dx$$

NOTE. $E[g(X)] \neq g(E[X])$

Percentile

Let X be a continuous random variable and let $0 \leq p \leq 1$. The p^{th} quantile or $100p^{th}$ percentile of the distribution of X is the smallest number q_p s.t.

$$F(q_p) = P(X \leq q_p) = p$$

The median of a distribution is its 50^{th} percentile.

Continuous Uniform Distribution

A random variable X is uniformly distributed on (a,b) if the density function of X is

$$f(x) = \frac{1}{b-a} \text{ for } a < x < b$$

and 0, otherwise.

We write $X \sim \text{Unif}(a,b)$.

CDF $F_X(x) = \dfrac{x-a}{b-a}$, for $a < x < b$

E(x) $\int_a^b \frac{x}{b-a}\,dx = \frac{x^2}{2(b-a)}\Big|_a^b = \left(\frac{1}{\cancel{b-a}}\right)\left(\frac{\overset{b+a}{\cancel{b^2-a^2}}}{2}\right)$

$\frac{b+a}{2}$

$$= \frac{b+a}{2}$$

V(x) $= \underbrace{E[X^2]} - \underbrace{\left(\frac{b+a}{2}\right)^2}_{E[X]^2}$

$\frac{(b-a)^2}{12}$

$$\int_a^b x^2 \left(\frac{1}{b-a}\right)dx = \frac{x^3}{3(b-a)}\Big|_a^b$$

$$= \frac{b^3-a^3}{3(b-a)} = \frac{\cancel{(b-a)}(b^2-ab+a^2)}{3\cancel{(b-a)}}$$

$$= \frac{b^2-ab+a^2}{3}$$

$$\frac{b^2-ab+a^2}{3} - \frac{b^2+2ab+a^2}{4} = \frac{(b-a)^2}{12}$$

Exponential Distribution

A continuous random variable has Exponential Distribution with parameter λ if its PDF f is given by $f(x)=0$ if $x<0$ and

$$f(x)=\lambda e^{-\lambda x} \quad \text{for } x \geq 0$$

We denote this distribution by $\text{Exp}(\lambda)$.

CDF	E(X)	V(X)	R
$F_x(x)=1-e^{-\lambda x}$ for $x \geq 0$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$	dexp(x, λ) pexp(x, λ)

Memoryless Property

A random variable X has the memoryless property if for all $0<s<t$,

$$P(X>t \mid X>s)=P(X>t-s)$$

$X \sim \text{Geo}(p)$

$F_x = 1-(1-p)^{x+1}$, $x=0,1,2\cdots$

$X \sim \text{Exp}(\lambda)$

$$F_x(x)=\begin{cases} 0, & x<0 \\ 1-e^{-\lambda x}, & x \geq 0 \end{cases}$$

Gamma Distribution

A continuous random variable has Gamma Distribution with parameters $\alpha > 0$ and $\lambda > 0$ if its PDF f is given by $f(x) = 0$ for $x > 0$ and

$$f(x) = \frac{\lambda(\lambda x)^{\alpha-1} e^{-\lambda x}}{\Gamma(\alpha)} \quad \text{for } x \geq 0,$$

Where the quantity $\Gamma(\alpha)$ is a normalizing constant such that f integrates to 1.

We denote this distribution by $\mathrm{Gam}(\alpha, \lambda)$.

Gamma function:

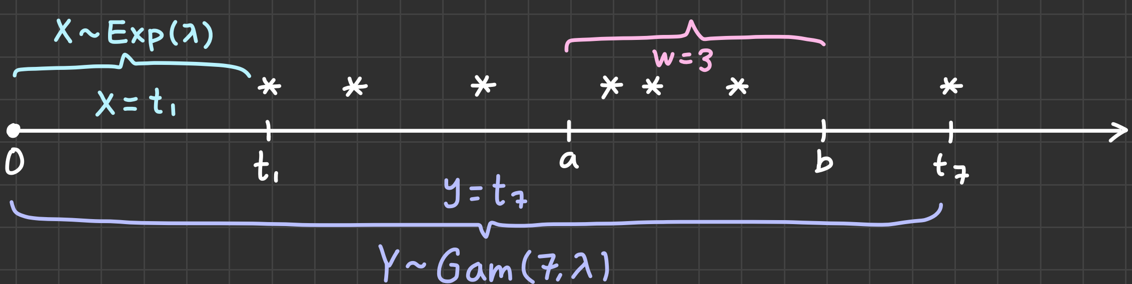
$$\Gamma(\alpha) = \int_0^\infty x^{\alpha-1} e^{-x} dx$$

1. For $\alpha > 1$, $\Gamma(\alpha) = (\alpha - 1)\Gamma(\alpha - 1)$
2. if $k \in \mathbb{N}^+$, $\Gamma(k) = (k-1)!$
3. $\Gamma(\frac{1}{2}) = \sqrt{\pi}$

$E(X)$	$V(X)$
$\frac{\alpha}{\lambda}$	$\frac{\alpha}{\lambda^2}$

$\mathcal{R}$ dgamma(x, α, λ)
pgamma(x, α, λ)

Poisson Process



Normal Distribution

A random variable X has the normal distribution with parameters μ and σ^2, if the density function of X is

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \quad -\infty < x < \infty$$

$$E(X) = \mu$$
$$Var(X) = \sigma^2$$

We write $X \sim Norm(\mu, \sigma^2)$ or $N(\mu, \sigma^2)$.

Standard Normal

$X \sim N(0,1)$

$$f_z(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2}, \quad -\infty < z < \infty$$

$$E(X) = 0$$
$$Var(X) = 1$$

$$F_z(z) = P(Z \le z) = \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2} dx$$
$$= \Phi(z)$$

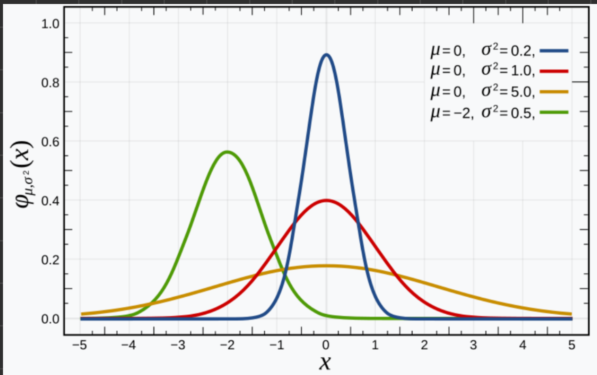
No closed form

R dnorm(x, μ, σ)
pnorm(x, μ, σ)

- Symmetric around $X = \mu$

ex) $P(X \le \mu) = P(X \ge \mu) = 0.5$

- Can convert any normal distribution to standard normal distribution with the transformation



$$Z = \frac{X-\mu}{\sigma}$$ (this is z-score which gives the number of σ X is away from μ)

$$P(X \le x) = P(Z\sigma + \mu \le x) = P\left(Z \le \frac{x-\mu}{\sigma}\right) = \Phi\left(\frac{x-\mu}{\sigma}\right)$$

The Empirical Rule for the Normal Distribution

$P(\mu-\sigma \leq X \leq \mu+\sigma) \approx 0.68$

$P(\mu-2\sigma \leq X \leq \mu+2\sigma) \approx 0.95$

$P(\mu-3\sigma \leq X \leq \mu+3\sigma) \approx 0.997$

Moment

$E(X^k)$ is called the k^{th} moment of X

Moment Generating Function

Let X be a random variable. The mgf of X is the real-valued function

$$m(t) = E[e^{tX}]$$

exists if $m_X(t)$ defined for an interval of t around 0 ($h>0$ s.t. $t \in (-h, h)$)

defined for all real t when this expectation exists.

Also written as $m_X(t)$.

$m_X(t) = E(e^{tX}) = \sum_{\forall x} e^{tx} p(x)$ ----- Discrete

$m_Y(t) = E(e^{tY}) = \int_{-\infty}^{+\infty} e^{ty} f(y)\,dy$ ----- Continuous

If mgf exists then:

- We can calculate the k^{th} moment of X as

$$E(X^k) = m_X^{(k)}(0)$$

- mgf uniquely specifies a probability distribution

- kth moment of X is $E(X^k) = \frac{d^k}{dt^k} m_X(t) \Big|_{t=0} = m_X^{(k)}(0)$

$$e^y = 1 + y + \frac{y^2}{2!} + \frac{y^3}{3!} + \cdots, \quad m_X(t) = E(e^{tX})$$

$$\therefore m_X(t) = E\left(1 + tX + \frac{(tX)^2}{2!} + \frac{(tX)^3}{3!} + \cdots\right)$$

$$= 1 + tE[X] + t^2 E\left[\frac{X^2}{2!}\right] + t^3 E\left[\frac{X^3}{3!}\right] + \cdots$$

Properties of mgf:

1. k^{th} moment of X as $E(X^k) = m_X^{(k)}(0)$
2. If X and Y are independent, then $m_{X+Y}(t) = m_X(t) m_Y(t)$
3. For constants $a \neq 0$ and b, $m_{aX+b}(t) = e^{bt} m_X(at)$
4. If two random variables have the same mgf, they have the same distribution.

$X \sim Bin(n, p)$	$m_X(t) = (pe^t + (1-p))^n$
$X \sim Pois(\lambda)$	$m_X(t) = e^{\lambda(e^t - 1)}$
$X \sim Geo(p)$	$m_X(t) = \frac{p}{1 - e^t(1-p)}$ Counting fails; $\frac{pe^t}{1 - e^t(1-p)}$ Counting all
$X \sim NB(r, p)$	$m_X(t) = \left(\frac{p}{1 - e^t(1-p)}\right)^r$
$X \sim Unif(a, b)$	$m_X(t) = \frac{e^{bt} - e^{at}}{t(b-a)}$
$X \sim Exp(\lambda)$	$m_X(t) = \frac{\lambda}{\lambda - t}, \ t < \lambda$
$X \sim Gam(\alpha, \lambda)$	$m_X(t) = \left(\frac{\lambda}{\lambda - t}\right)^\alpha, \ t < \lambda$
$X \sim N(\mu, \sigma^2)$	$m_X(t) = e^{\mu t + \sigma^2 t^2 / 2}$

Common Discrete Probability Distribution Table

Name	$X =$	Formula $P(X=k) =$	$E(X)$	$V(X)$
Binomial $X \sim Bin(n, p)$	# of success in n trials	$\binom{n}{k} p^k (1-p)^{n-k}$ for $k = 0, 1, 2, \cdots, n$	np	$np(1-p)$
Poisson $X \sim Pois(\lambda)$	# occurrences of event in a fixed interval	$\frac{e^{-\lambda}\lambda^k}{k!}$ for $k = 0, 1, 2, \cdots$ and $\lambda > 0$	λ	λ
Hypergeometric $X \sim HyperGeo(r, N, n)$ $\mathcal{R}$: · hyper(k, r, N-r, n)	# of successes in sample of size n	$\frac{\binom{r}{k}\binom{N-r}{n-k}}{\binom{N}{n}}$ for $\max(0, n-(N-r)) \le x \le \min(n, r)$	$n\frac{r}{N}$	$n\left(\frac{N-n}{N-1}\right)\frac{r}{N}\left(1-\frac{r}{N}\right)$
Geometric $X \sim Geo(p)$	# of failures before first success	$p(1-p)^k$ for $k = 0, 1, 2, \cdots$ * cdf: $1-(1-p)^{k+1}$	$\frac{1-p}{p}$	$\frac{1-p}{p^2}$
Negative Binomial $X \sim NB(r, p)$	# of failures before r^{th} success	$\binom{k+r-1}{r-1} p^r (1-p)^k$ for $k = 0, 1, 2, \cdots$	$\frac{r(1-p)}{p}$	$\frac{r(1-p)}{p^2}$

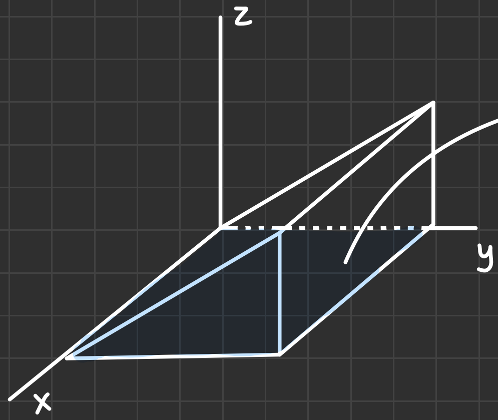
Common Continuous Probability Distribution Table

Name	Formula $f_X(X=x)=$	$E(X)$	$V(X)$
Uniform $X \sim Unif(a,b)$	$f(x)=\begin{cases} \frac{1}{b-a}, & a<x<b \\ 0, & \text{otherwise} \end{cases}$	$\frac{b+a}{2}$	$\frac{(b-a)^2}{12}$
Exponential $X \sim Exp(\lambda)$	$f(x)=\begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & \text{otherwise} \end{cases}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$
Gamma $X \sim Gam(\alpha,\lambda)$	$f(x)=\begin{cases} \frac{\lambda(\lambda x)^{\alpha-1} e^{-\lambda x}}{\Gamma(\alpha)}, & x \geq 0 \\ 0, & \text{otherwise} \end{cases}$	$\frac{\alpha}{\lambda}$	$\frac{\alpha}{\lambda^2}$
Normal $X \sim N(\mu,\sigma^2)$	$f(x)=\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \quad -\infty < x < \infty$	μ	σ^2

Discrete vs. Continuous

	Discrete	Continuous
probability distribution	pmf $p(x)$ probability mass function	pdf $f(x)$ probability density function
PMF/PDF	$p(x) \geq 0$ for all x adds to 1	$f(x) \geq 0$ for all x area under curve = 1
CDF	$F(y) = \sum_{x=-\infty}^{y} p(x)$	$F(y) = \int_{-\infty}^{y} f(x)\,dx$
$P(a \leq X \leq b)$	$F(b-1) - F(a-1)$	$\int_a^b f(x)\,dx = F(b) - F(a)$
$E[h(X)]$	$\sum_{x=-\infty}^{\infty} h(x)p(x)$	$\int_{-\infty}^{\infty} h(x)f(x)\,dx$

	Discrete RV	Continuous RV
Joint pdf/pmf	$P_{X,Y}(x,y) = P(X=x)P(Y=y)$ if X and Y independent $\sum_{x\in S}\sum_{x\in T} P(X=x, Y=y) = 1$ $P(a\le X\le b, c\le Y\le d) = \sum_{x=a}^{b}\sum_{x=c}^{d} P(X=x, Y=y)$	$f(x,y) = f_X(x)f_Y(y)$ if X and Y independent $\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\,dx\,dy = 1$ $S\subseteq \mathbb{R}^2,\ P((X,Y)\in S) = \iint_S f(x,y)\,dx\,dy$
Marginal pdf/pmf	$p_X(x) = \sum_{\text{all } y} p_{X,Y}(x,y)$ $p_Y(y) = \sum_{\text{all } x} p_{X,Y}(x,y)$	$f_X(x) = \int_{-\infty}^{\infty} f(x,y)\,dy$ → range of y $f_Y(y) = \int_{-\infty}^{\infty} f(x,y)\,dx$ → range of x
Joint cdf	$F(a,b) = P(X\le a, Y\le b)$ $= \sum_{a\le x}\sum_{b\le y} p_{X,Y}(a,b)$	$F(x,y) = P(X\le x, Y\le y)$ $= \int_{-\infty}^{x}\int_{-\infty}^{y} f(s,t)\,dt\,ds$
E[X]	$\sum_{\text{all } y}\sum_{\text{all } x} x\,p_{X,Y}(x,y)$	$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} x f(x,y)\,dx\,dy$
E[Y]	$\sum_{\text{all } y}\sum_{\text{all } x} y\,p_{X,Y}(x,y)$	$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} y f(x,y)\,dx\,dy$
E[g(X,Y)]	$\sum_i\sum_j g(a_i,b_j)P(X=a_i, Y=b_j)$	$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} g(x,y) f(x,y)\,dx\,dy$
X | Y Conditional	$p_{X\mid Y}(x\mid y) = \dfrac{p_{X,Y}(x,y)}{p_Y(y)}$	$f_{X\mid Y}(x\mid y) = \dfrac{f_{X,Y}(x,y)}{f_Y(y)}$
E[X | Y=y]	$\sum_{\text{all } x} x\,p_{X\mid Y}(x\mid y)$	$\int_{-\infty}^{\infty} x f_{X\mid Y}(x\mid y)\,dx$



Region of (x,y) s.t $f_{X,Y}(x,y) > 0$ is called Support

How to find the density of $Y = g(x)$

1. Determine possible values of Y based on the value of X and $g(x)$.
2. Begin with cdf $F_Y(y) = P(Y \le y) = P(g(X) \le y)$. express the cdf in terms of X.
3. From $P(g(X) \le y)$ obtain an expression $P(X \le \cdots)$. right side will be function of y.
   - If g is invertible, $P(g(X) \le y) = P(X \le g^{-1}(y))$.
4. Differentiate, with respect to y to obtain $f_Y(y)$.

Two-dimensional change-of-variable formula

Let X and Y be random variables.
Let $g: \mathbb{R}^2 \to \mathbb{R}^2$ be a function.

If X and Y are discrete RVs with values $a_1, a_2, \cdots$ and $b_1, b_2 \cdots$

$$E[g(X,Y)]$$

$$= \sum_i \sum_j g(a_i, b_j) P(X = a_i, Y = b_j)$$

If X and Y are continuous RVs with joint pdf f

$$E[g(X,Y)]$$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x,y) f(x,y)\, dx\, dy$$

Conditional pmf of two-variable

The conditional pmfs of Y given X, and X given Y where X and Y are discrete RVs are:

$$p_{Y|X}(y|x) = \frac{p_{X,Y}(x,y)}{p_X(x)} \qquad \Big| \qquad p_{X|Y}(x|y) = \frac{p_{Y|X}(y,x)}{p_Y(y)}$$

Independent Random Variables

Two RVs X and Y are independent if for every pair of (x, y) values,

$$p_{X,Y}(x,y) = p_X(x)\,p_Y(y) \text{ for discrete X and Y}$$

$$f_{X,Y}(x,y) = f_X(x)\,f_Y(y) \text{ for continuous X and Y}$$

If this is not true for at least one (x,y), RVs X and Y are independent.

Covariance

Let X and Y be jointly distributed discrete / continuous random variables and let $\mu_X = E(X)$ and $\mu_Y = E(Y)$.

The covariance of X and Y is

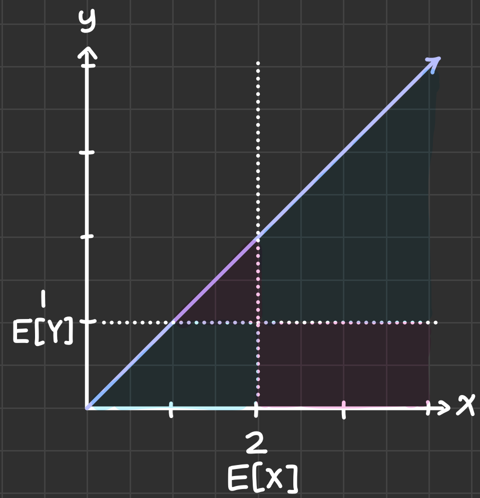
$$\text{Cov}(X,Y) = E[(X-\mu_X)(Y-\mu_Y)]$$
$$= E(XY) - E(X)E(Y)$$

inverse not always true

if X > mean when Y > mean
$\text{Cov}(X,Y) > 0$

if X > mean when Y < mean
$\text{Cov}(X,Y) < 0$

if X and Y are independent
$\text{Cov}(X,Y) = 0$



- Assume $E(X)=2,\ E(Y)=1$
- If Area > Area

  $\rightarrow Cov(X,Y) > 0$
- If Area > Area

  $\rightarrow Cov(X,Y) < 0$

Variants of Difference between RV

- $V[X-Y] = E[(X-Y)^2] - E[X-Y]^2$

$$= E[X^2+2XY+Y^2] - (E[X]-E[Y])^2$$

$$= E[X^2] + 2E[XY] + E[Y^2] - (E[X]^2 - 2E[X]E[Y] + E[Y]^2)$$

$$= (E[X^2]-E[X]^2) + (E[Y^2]-E[Y]^2) - 2(E[XY]-E[X]E[Y])$$

$$= V[X] + V[Y] - 2Cov(X,Y)$$

For two RV X, Y, and constants a, b, c:

$$E[aX+bY+c] = aE[X] + bE[Y] + c$$

$$V[aX+bY+c] = a^2V[X] + b^2V[Y] + 2ab\,Cov(X,Y)$$

$\hookrightarrow V[a_1X_1 + a_2X_2 + \cdots + a_nX_n] = \sum_{i=1}^{n}\sum_{j=1}^{n} a_ia_j Cov(X_i, X_j)$

Note $Cov(X,X) = V[X]$

Correlation

Let X and Y be two RVs.
The correlation coefficient $\rho(x,y)$ is defined to be 0 if $V[X]=0$ or $V[Y]=0$. otherwise:

$$\rho(X,Y) = \frac{Cov(X,Y)}{\sqrt{V[X]V[Y]}}$$

- If X and Y are independent, $\rho(x,y)=0$.
  (However, $\rho(x,y)=0 \not\Rightarrow$ X and Y are independent)
- For any two RVs, $-1 \leq \rho(x,y) \leq 1$. * magnitude: strength of association
- Iff $\rho(X,Y)=\pm 1$, $Y=aX+b$ for some $a \neq 0$.
- If a and c are both positive / negative:

$$Cov(aX+b, cY+d) = acCov(X,Y)$$
$$\rho(aX+b, cY+d) = \rho(X,Y)$$

Statistics ⟶ is a random variable!

A statistic is any quality that could be calculated from sample data.

Uppercase represents statistic, a RV. — X_i — i^{th} randomly selected unit

- A probability distribution of a statistic is often called its sampling distribution.

The RVs $X_1, X_2, \cdots X_n$ forms a (simple) random sample of size n if:
1. X_i's are independent
2. Every X_i has same prob. distribution

also referred as i.i.d

Weak Law of Large Numbers

Let $X_1, X_2, \cdots$ be an i.i.d. sequence of RVs with finite mean μ and variance σ^2.
For $n = 1, 2, \cdots$, let $S_n = X_1 + \cdots + X_n$. Then:

$$P\left(\left|\frac{S_n}{n} - \mu\right| \geq \varepsilon\right) \to 0$$

for all $\varepsilon > 0$

$\bar{X}_n$ converges in probability to $E[X] = \mu$

Strong Law of Large Numbers

(Same assumptions)

$$P\left(\lim_{n\to\infty} \frac{S_n}{n} = \mu\right) = 1$$

$\bar{X}_n = \frac{S_n}{n}$ (mean of n RVs)

$$\mu_{\bar{x}} = E(\bar{x}_n) = E\left(\frac{\sum_{i=1}^n X_i}{n}\right)$$

$$= E\left(\frac{X_1 + X_2 + \cdots + X_n}{n}\right) = \frac{1}{n}\sum_{i=1}^n \overbrace{E[X_i]}^{\mu_x} = \frac{\cancel{n}\mu_x}{\cancel{n}} = n$$

$$\sigma^2_{\bar{x}} = V(\bar{x}_n) = V\left(\frac{\sum_{i=1}^n X_i}{n}\right)$$

$$= V\left(\frac{X_1 + X_2 + \cdots + X_n}{n}\right) = \frac{1}{n^2}\sum_{i=1}^n \overbrace{V[X_i]}^{\sigma_x^2} = \frac{\cancel{n}\sigma_x^2}{n^{\cancel{2}}} = \frac{\sigma_x^2}{n}$$

$V[aX + bY + c]$
$= a^2V[X] + b^2V[Y] + \cancel{2ab\,Cov(X,Y)}^{\,0}$

$Cov(X,Y) = 0$ if X, Y are independent

Central Limit Theorem

Let $X_1, X_2, \cdots$ be an i.i.d. sequence of RVs with finite mean μ and variance σ^2.
For $n = 1, 2, \cdots$, let $S_n = X_1 + \cdots + X_n$.

Then the distribution of standardized RV $\frac{S_n/n-\mu}{\sigma/\sqrt{n}}$ converges to a standard normal distribution. For all t,

$$P\left(\frac{S_n/n-\mu}{\sigma/\sqrt{n}} \le t\right) \rightarrow P(Z \le t), \text{ as } n \to \infty.$$

$Z \sim Norm(0,1)$

Important Theorems we might need

TRANSFORMATION THEOREM Let X have pdf $f_X(x)$ and let $Y = g(X)$, where g is monotonic (either strictly increasing or strictly decreasing) on the set of all possible values of X, so it has an inverse function $X = g^{-1}(Y) = h(Y)$. Assume that h has a derivative $h'(y)$. Then

$$f_Y(y) = f_X(h(y)) \cdot |h'(y)| \tag{4.12}$$

PROPOSITION Let $X_1, X_2, \ldots, X_n$ be a random sample from a distribution with mean value μ and standard deviation σ. Then

1. $E(T_o) = n\mu$
2. $V(T_o) = n\sigma^2$ and $\sigma_{T_o} = \sqrt{n}\sigma$
3. If the X_i's are normally distributed, then T_o is also normally distributed.

1. $E(\overline{X}) = \mu$
2. $V(\overline{X}) = \frac{\sigma^2}{n}$ and $\sigma_{\overline{X}} = \frac{\sigma}{\sqrt{n}}$
3. If the X_i's are normally distributed, then $\overline{X}$ is also normally distributed.